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Advances on methane steam reforming to produce hydrogen through membrane reactors technology: A review

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ABSTRACT
Methane steam reforming is the most common industrial process used for almost the 50% of the world's hydrogen production. Commonly, this reaction is performed in fixed bed reactors and several stages are needed for separating hydrogen with the desired purity. The membrane reactors represent a valid alternative to the fixed bed reactors, by combining the reforming reaction for producing hydrogen and its separation in only one stage. This article deals with the recent progress on methane steam reforming reaction, giving a short overview on catalysts utilization as well as on the fundamentals of membrane reactors, also summarizing the relevant advancements in this field.

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1. Introduction

Almost 50% of the world-wide demand for hydrogen (the estimated production of hydrogen is ~ $55{\cdot}10^6$ tons/year) is derived from natural gas (1). In particular, the current conventional methods for generating hydrogen from natural gas include steam reforming, partial oxidation and autothermal reforming.

Natural gas steam reforming, called as methane steam reforming (MSR) owing to its main component, represents the most common process for hydrogen generation, despite its negative impact on the climate through emissions of greenhouse gases (GHGs) (2). Today, approximately 50% of the global demand for hydrogen is provided by this process, while nearly 30% derives from oil/naphtha reforming in refinery/chemical industrial off-gases, 18% from coal gasification, 3.9% from water electrolysis, and 0.1% from other sources (3).

Currently, the conversion of natural gas (mainly composed by methane) to hydrogen is carried out in a conventional reformer (CR) that is followed by water-gas shift (WGS) reactors (high temperature and low temperature WGS

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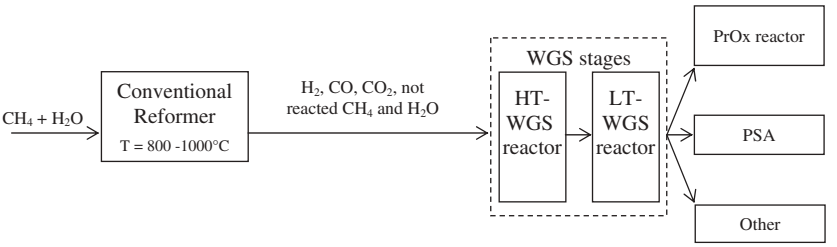


Figure 1. Scheme of the conventional multi-stages process for high-grade hydrogen production via MSR reaction.

reactors) and equipment for H_2 separation and purification (pressure swing adsorption (PSA), preferential oxidation reactors (PrOx), etc.). A schematic view of the aforementioned multi-steps process is illustrated in Fig. 1.

Methane and steam are supplied to the CR where they react (Eqs. (1)–(3)) under harsh operating conditions, namely T = 800–1000°C and 14–20 bar over Ni-based catalysts.

$$CH_4 + H_2O = CO + 3H_2 \qquad \Delta H^0_{298K} = 206\,kJ/mol \tag{1}$$

$$CH_4 + 2H_2O = CO_2 + 4H_2 \qquad \Delta H^0_{298K} = 165\,kJ/mol \tag{2}$$

$$CO + H_2O = CO_2 + H_2 \qquad \Delta H^0_{298K} = -41\,kJ/mol \tag{3}$$

The reforming process produces a hydrogen-rich gas mixture with significant CO content (generally higher than 5 vol% (4)). Consequently, two WGS reactors arranged in series are used to reduce the CO content (Eq. (3)) of the reformed stream and, at the same time, increase the hydrogen content. Typically, the first WGS reactor (HT-WGS) is loaded with a high temperature catalyst to favor rapid kinetics, resulting in high equilibrium CO selectivity and decreased hydrogen yield. The HT-WGS reactor operates at temperatures between 310 and 450°C using such catalysts as Fe_3O_4/Cr_2O_3. In this stage, the content of CO concentration is decreased by 50% (5,6). The second WGS reactor (LT-WGS) is loaded with low-temperature catalysts such as $Cu/ZnO/Al_2O_3$-based and operates between 180 and 250°C (6–8). In this reactor, the CO concentration is decreased from 3vol% to 500 ppm.

Therefore, other techniques have to be adopted for further purifying hydrogen and the method chosen depends on the hydrogen purity requirements for the specific process (Table 1). As illustrated in Fig. 1, the PrOx reactor represents one of the different solutions useful to produce a highly concentrated hydrogen stream. Commonly, this reactor is loaded with catalysts such as Pt, Ru, and Rh supported on Al_2O_3, in which the selective oxidation of CO into CO_2 takes place. The PrOx reactor shows both benefits and drawbacks. In particular, it adds complexity to the whole system owing

Table 1. Various techniques for H_2 purification.

Technique	Description	Drawbacks	Benefits
Preferential oxidation	Selective oxidation of CO into CO_2	Added complexity to the whole system	CO concentration of the final H_2 rich-stream around 500 ppm
PSA	Selective adsorption of compounds from a gas stream	Around 20% of H_2 is lost under operation (H_2 recovery between 70 and 85%)	H_2 purity > 99.99%
Methanation	Methanation reactors makes possible the CO content reduction	Unnecessary consumption of hydrogen	Methanation reactors is simpler and no addition of air is required
Cryogenic distillation	Partial condensation of gas mixtures at low temperatures	Relatively low H_2 purity between 90 and 98%	Low operating temperature H_2 recovery up to 98%
Dense palladium membrane	Selective diffusion of hydrogen through a palladium alloy membrane	High cost and low mechanical resistance	H_2 recovery up to 99% H_2 purity > 99.999%
Polymeric membrane	Differential rate of diffusion of gases through a permeable membrane	Relatively low H_2 purity between 92–98	Low cost H_2 recovery > 85%
Metal hybride separation	Reversible reaction of hydrogen with metals to form hydrides	H_2 recovery between 75 and 95%	H_2 purity ˜ 99
Solid polymer electrolyte cell	Electrolytic passage of hydrogen ions across a solid polymer membrane	Sluggish anode reaction coupled with the inefficient cathode reaction (low overall cell performance)	H_2 recovery ˜ 95% H_2 purity ˜ 99.8%

to the air supplying, although the H_2-rich stream contains only 500 ppm of CO.

Alternatively, hydrogen can be purified by other techniques, namely PSA, cryogenic distillation, methanation, physical scrubbing (for separating initially CO_2), membrane technologies, etc., which can ensure the required hydrogen purity.

As summarized in Table 1, cryogenic distillation does not allow for significant hydrogen separation/purification, although the operating temperature is low (9). The technology based on Pd-membranes makes possible to reach high hydrogen purity and recovery, despite some challenges needs to be overcome such as high membrane cost, low mechanical resistance as well as low tolerance to sulphuric compounds and unsaturated hydrocarbons (10).

Regarding to polymeric membranes utilization, their limited thermal and mechanical resistance, relatively high sensitivity to swelling and compaction, low tolerance to hydrochloric acid (HCl) and sulfur oxides (SO_x) make polymeric membranes-based processes less attractive.

Currently, the PSA process is the most used conventional technique, allowing for high hydrogen purity. Nevertheless, a loss of hydrogen nearly 20% under PSA operation represents the main drawback of this process.

Although at both academic and industrial level MSR represents a well-established process, many studies are still ongoing. Nowadays, the research on the MSR reaction is mainly devoted to enhance the catalyst performance by producing as much hydrogen as possible, while emphasizing the resistance to catalyst sintering and minimizing the effect of the carbon deposition and sulphur poisoning. Recently, a number of studies is focused on the experimental evaluation of both noble and non-noble metals-based catalysts and the effect of the metal-support interactions (11–14). Among them, a special case is represented by Ni-based catalysts due to their low cost besides other positive characteristics (15–19).

Furthermore, in the last two decades, numerous researchers studied MSR reaction, investigating the benefits in the use of alternative solutions to the CRs such as of membrane reactors (MRs) (20–33) and, more recently, fluidized-bed membrane reactors (FBMRs) (34–41). Another emerging field of interest is also represented by micro and macro MRs. In particular, the micro-membrane reactors (MMRs), distinguished in planar microchannel, hollow-fiber, and monolithic configurations, represent a very promising area from both a scientific and technological point of view owing to their potential application as on-board fuel reformers for miniaturized PEMFCs or, more generally, for portable electronic devices (42–51).

Most recently, growing attention has been given to hydrogen generation from various renewable and bio-derived sources, particularly from biogas, in which methane is the main product (52–67). Indeed, biogas is a versatile raw

feedstock for reforming processes and can represent an alternative to natural gas exploitation, thereby reducing the GHGs emissions. The composition of biogas can vary greatly, depending on which kind of residual biomass among animal waste, sewage treatment plants or industrial wastewater, landfills, etc. is used during the anaerobic digestion process. However, biogas is mainly composed of methane and carbon dioxide, in addition to traces of H_2S, NH_3, hydrogen, nitrogen, oxygen and steam (68). In terms of the compound distribution, methane can vary between 55% and 70%, carbon dioxide between 30% and 45%, H_2S from 500–4000 ppm, NH_3 from 100–800 ppm, while hydrogen, nitrogen, oxygen and steam can show percentage lower than 1% (vol.) (53,54).

In this article, the recent progress regarding the most relevant topics of MSR reaction via MRs technology are described and critically discussed. Furthermore, a short overview on the mathematical models used for modelling and simulating the MSR reaction in MRs is also illustrated.

2. A short overview on the catalysts for methane steam reforming reaction

Although the aim of this work is focused on the recent advances regarding the MSR reaction in MRs, in this section a short overview on the traditional catalysts and the recent developments in this area are summarized.

In particular, the research done in the last years on MSR reaction demonstrated that most used catalysts show low tolerance to poisons requiring special activation procedures (69). More recently, some studies have been also devoted to the development of new catalysts, that are able to overcome issues related to sintering and coke formation resistance, to allow for hydrogen production near the theoretical limit, to improve stability and reduce the energy input.

The most investigated catalysts in this field are characterized by single noble and non-noble metal belonging to the VIII group (15,70–81). Currently, it's not yet possible to give a specific order of catalytic activity and selectivity towards hydrogen production via MSR reaction owing to the disagreement among several scientists involved in this field. Indeed, Jones et al. (82) proposed the following catalytic activity order: Ru > Rh > Ir > Pt, indicating the Pt catalyst as the less active among the other metals. From the opposite side, the study of Wei and Iglesia (83) showed that Pt performance in terms of C-H bond activation are higher than to Ir, Rh, and Ru.

However, although all these noble catalysts show high catalytic activity and less coke formation, the nickel catalyst represents the most studied and commonly used at industrial scale for MSR reaction owing to its low cost. (70). Figure 2 shows a qualitative percentage distribution of the main analysed catalysts during MSR reaction where the numerous investigations

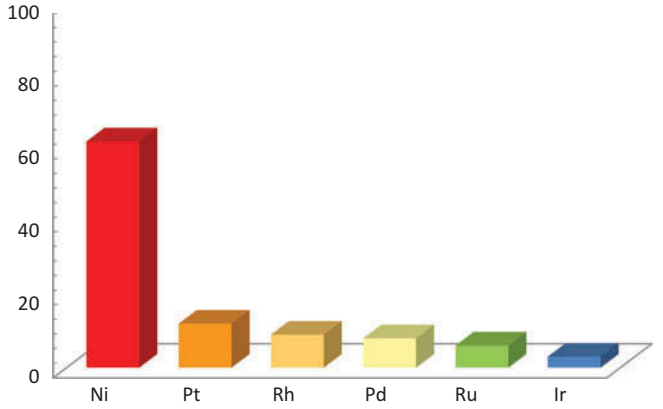


Figure 2. Qualitative distribution about catalysts utilization for MSR reaction.

carried out on Ni catalyst are glaring. In particular, a large number of studies is addressed toward the improvement of Ni performance, by analyzing the effect of different supports and promoters (84–88). Recently, Nieva et al. (15) stated that the catalytic activity of Ni-based catalysts mainly depends on the support, which plays an important role in the catalytic process. In fact, it is responsible for the metal dispersion, affecting the resistance to sintering and, sometimes, directly participating to the reaction by facilitating the reactants adsorption. Furthermore, the same authors demonstrated that, during MSR reaction at 600°C, the activity order of Ni-based catalysts is influenced by different supports as in the following order: $Ni/MgAl_2O_4 > Ni/ZnAl_2O_4 > Ni/Al_2O_3 > Ni/SiO_2$. In particular, Ni/SiO_2 seems to be deactivated very quickly owing to the surface oxidation and carbon deposition, whereas $Ni/ZnAl_2O_4$ shows the lowest carbon deposition and a high resistance to sintering.

In other studies, it was observed that such supports as Al_2O_3 and SiO_2 allow for a gradual oxidization of Ni-catalysts during MSR reaction at 500°C and ambient pressure, whereas the utilization of ZrO_2 makes stable the Ni-catalyst and allows for higher methane conversion than other aforementioned supports (86). Indeed, the ZrO_2 support allows a water accumulation, favouring the formation of hydroxyl groups promoting MSR reaction.

Considering its high thermal and mechanical resistance as well as high oxygen storage capacity, CeO_2 has been mostly studied as support or promoter of Ni-catalysts (84,85,87–92). In particular, Dan et al. (87) studied MSR reaction over Ni-catalysts deposited on Al_2O_3 support, which was previously modified with CeO_2 and La_2O_3. They observed that the morphological characteristics, in terms of surface area and nickel dispersion, make possible an enhancement of the catalytic properties with a consequent increase of the methane conversion and a further decrease of coke formation.

As reported above, there is a comprehensive literature about catalysts utilized for MSR reaction based on single metal-phase on activated supports.

Nevertheless, in recent years a more current approach for improving the catalyst performance in terms of hydrogen production comes from the utilization of bi-metallic catalysts (69). Indeed, the combination of two metals can be more effective than the single metal-based catalysts in terms of enhanced conversions and hydrogen selectivity, besides resistance to coke formation and higher catalytic activity. On the contrary, the preparation and characterization of this new class of catalysts seem to be more complicated owing to the presence of different metals on the support. As a consequence, many studies have been addressed on the metal combinations (15,87,93–106) and Table 2 summarizes some of the most recent studies present in the open literature. In various cases reported in this table, the characteristics of the considered catalysts indicate "high stability," meaning stable performance of the catalyst in terms of conversion and hydrogen selectivity for relatively long period, meanwhile observing high resistance to sintering effect.

As reported by various authors on this topic, in case of bimetallic catalysts utilization, the two metals do not show a deep interaction with each other, but it seems that one of them acts as a dopant. Indeed, in several cases the depletion of coke deposition is attributed to the interaction of a metal acting as a dopant on the other metal and preventing the formation of carbon-bonds with the latter. Furthermore, the "dopant-like" effect of one of the two metals combined to the role of the support influences globally the activity of the catalyst towards MSR reaction (11).

3. Fundamentals of Pd-based membranes

3.1. *Membrane and membrane reactor technology in the market*

In recent decades, a wide scientific literature has been addressed towards the study of inorganic membranes, particularly, dense metallic membranes because of their benefits in the gas separation and applications in MRs (10,107,108). In particular, Pd-based membranes have received particular attention owing to their capability to separate hydrogen from other gas, although their large commercialization is still limited because some challenges need to be overcome as the low hydrogen permeability, mechanical resistance, and high costs.

However, at pilot/industrial scale, one the first commercial applications regarding dense Pd-membranes applied to hydrogen rich-gas purification was realized by Johnson Mattey in 1964, obtaining high grade hydrogen with 0.1 ppm of impurities. Successively, they used this technology for realizing a MR and carrying out methanol steam reforming reaction to generate hydrogen for the Britsh Antarctic Survey.

In Japan, the first pilot scale Pd-based MR for pure hydrogen generation was developed by Tokyo Gas Company Ltd. for carrying out MSR reaction

Table 2. Bimetallic catalysts for methane reforming in conventional reactors.

Bimetallic catalysts	Characateristics	Operating temperature [°C]	Reference
Ni-Zn	Low carbon deposition and high CO selectivity at medium-high temperature	600	Nieva et al. (15)
Ni-Ag	Improved methane conversion and absence of carbon deposition	> 500	Dan et al. (87)
Ni-Mo	Prepared using co-impregnation method, it keeps very stable performance and low carbon deposition	800	Huang et al. (93)
Pd-Rh	High stability and better performance than Ni and Ru-based catalyst with high conversion and H_2 yield	850	Roy et al. (94)
Ni-Mg	High activity and very low carbon deposition	800	Takeguchi et al. (95)
Pt-Ni	High stability and fast start-up/shut-down	700	Li et al. (96)
Ni-La	High stability and low carbon formation	850	Hufschmidt et al. (97)
Ru-Ni	Due to the Ni-Ru interactions the catalytic performances are improved	800	Nawfal et al. (98)
Ni-Si	11% of Ni nanoparticles with Si core in core-shell structure	750	Majewski et al (99)
Ni-Si	High stability with 10% of Ni content	700	Bej et al. (100)
Rh-Sr	High stability and activity	> 700	McGuire et al. (101)
Ni-Mo	High stability and low carbon formation	700	Maluf and Assaf (102)
Ni-B	Improved stability related to boron presence	700–800	Xu et al. (103)
Ce-Gd	High resistance to coke deposition and high temperatures	> 850	Ramírez-Cabrera et al. (104)
Ni-Au	Absence of carbon deposition	550	Chin et al. (105)
Ni-Cu	Improved methane conversion and high resistance to coke	550	Khzouz et al. (106)

by supplying "city gas" (88.5% of methane, 4.6% of ethane, 5.4% of propane, and 1.5% of butane). The MR allows for the production of ultrapure hydrogen for 40 Nm^3/h (hydrogen concentration of the permeate stream equal to 99.9999%) (88).

Nowadays, various companies sell membrane-based systems for separating hydrogen at pilot scale and, among them, Air Products, Grasys, IGS, Hysep, SepMem, DivexGlobal, and Air Liquide Industries are worth of noting, even though not all of them utilize Pd-based membrane technology (see Table 3).

3.2. *Hydrogen transport in Pd-based membranes*

As is well known, the hydrogen molecular transport through dense, self-supported Pd-based membranes occurs via solution/diffusion mechanism, which involves six different stages (107): (1) the dissociation of molecular hydrogen at the gas/metal interface; (2) adsorption of the atomic hydrogen on the membrane surface; (3) dissolution of atomic hydrogen into the palladium matrix; (4) diffusion of atomic hydrogen through the membrane; (5) re-combination of atomic hydrogen to form hydrogen molecules at the gas/metal interface; and (6) desorption of hydrogen molecules.

The hydrogen flux permeating through the membrane can be expressed by the Sieverts-Fick's law as in the following (Eq. (4)):

$$J_{H_2,Sieverts-\mathrm{Fick}} = Pe_{H_2} \cdot \left({p^{0.5}}_{H_2,retentate} - {p^{0.5}}_{H_2,permeate} \right) / \delta, \tag{4}$$

where $J_{H2,Sieverts\text{-}Fick}$ represents the hydrogen flux permeating through the dense Pd-based membrane, Pe_{H2} the hydrogen permeability, δ the membrane thickness, $p_{H2\text{-}retentate}$ and $p_{H2\text{-}permeate}$ the hydrogen partial pressures in the retentate (reaction side) and permeate (side in which hydrogen permeating through the membrane is collected) zones, respectively, and the "0.5" coefficient is the dependence factor of the hydrogen flux to the hydrogen partial pressure.

The dependence of the hydrogen permeability on the temperature can be expressed by an Arrhenius-like equation (Eq. (5)):

$$Pe_{H_2} = {Pe^0}_{H_2} \, exp(-\mathrm{E_a}/\mathrm{RT}), \tag{5}$$

where Pe^0 represents the pre-exponential factor, E_a the apparent activation energy, R the universal gas constant, and T the absolute temperature.

Consequently, the Arrhenius like Eq. (5) can be substituted in the Sieverts-Fick's law, globally representing the Richardson's equation (Eq. (6)):

$$J_{H_2} = {Pe^0}_{H_2}(exp(-\mathrm{E_a}/\mathrm{RT})) \cdot \left({p^{0.5}}_{H_2,retentate} - {p^{0.5}}_{H_2,permeate} \right) / \delta. \tag{6}$$

Table 3. Industrial companies using membrane-based technology for hydrogen separation/purification.

Separation typology	p_{FEED} (bar)	Expected H_2-recovery [%]	$H_{2\text{-PERMEATE PURITY}}$ [%]	T [°C]	Membrane	Company
H_2/CO; H_2/CO_2, etc.	138	99	99,9		(Carbon steel - 6-40-P1-HP-CS-H_2)	IGS
High grade H_2 recovery	10–120	–	90–99.5	5–40	CarboPEEK	GRASYS
High grade H_2 recovery	2–70	–	99.5–99.995	300–450	Pd-based	HYSEP
High grade H_2 recovery	15–80	70–97	90–95	–	Polymeric	UOP – A Honeywell Company
High grade H_2 recovery	–	90–95	90 – 99.9	–	–	SEPMEM
H_2/CH_4	30	92,6	92,3	25	–	AIR LIQUID
N_2,O_2,Ar removal from H_2 rich-gas	–	95[a]	99	–	Helipur	DIVEX GLOBAL

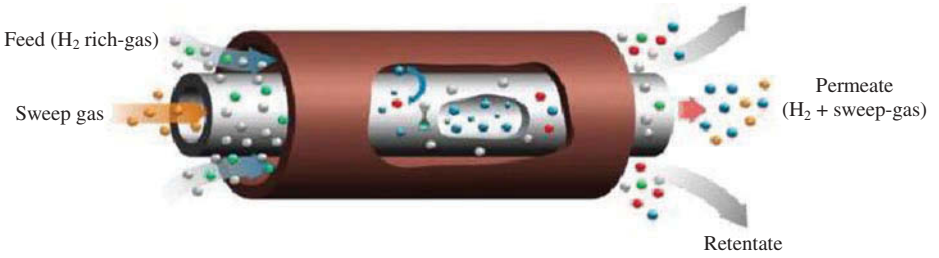


Figure 3. Scheme of a tubular dense Pd-based permeation module for H2 separation.

In case of validity of Sieverts-Fick's law, the dense Pd-based membrane is fully hydrogen perm-selective with respect to all the other gases, which means that only hydrogen can permeate through the membrane; see Figure 3.

Equation (6) points out the inverse proportionality of the hydrogen permeating flux to the membrane thickness. In particular, the role played by the membrane thickness is relevant because thinner membranes allow for higher permeability, whereas thicker membranes ensure the mechanical resistance (108).

As one of the main drawbacks of dense Pd-membranes is their susceptibility to cracking owing to the hydrogen absorbed, that causes the phase transition from α to β palladium hydride at temperatures lower than 300°C and pressures below 2.0 MPa. This effect is called "embrittlement phenomenon," in which the dissolved hydrogen makes possible the elongation of the metallic film (that produces the α ↔ β hydride transformations), leading to fractures after repeated pressure and thermal cycles. The embrittlement phenomenon can be avoided by alloying the Pd with metals such as silver, gold because the presence of a second metal expands the membrane lattice making it less influenced by the hydrogen permeation and, then, less brittle than the pristine Pd. In practice, the metal alloyed to palladium plays the role of an electron donor like the hydrogen in the Pd, allowing for the hydride formation in the Pd-alloyed membrane with a hydrogen concentration lower than pristine Pd membranes. Furthermore, the increased presence of a second metal alloyed to palladium favours an improvement of the hydrogen solubility, meanwhile lowering the diffusion coefficient of hydrogen through the membrane.

Drawbacks affecting both dense Pd and Pd-alloys membranes are represented by the contamination/deactivation of the palladium by carbon compounds at temperature higher than 450°C, the irreversible poisoning by sulphur compounds and the high cost of Pd.

On the contrary, porous membranes, such as aluminium, titanium, or silica oxides, are chemically inert, stable at high temperatures, and exhibit relatively high hydrogen permeability, resulting available at moderate cost. Nevertheless, they show poor performance in terms of hydrogen perm-selectivity and, in case of utilization in MR applications, a significant amount of reactants is lost for permeation through the membrane.

Composite membranes, based on a dense Pd-layer deposited onto a porous support, constitute a good compromise between dense Pd-based and porous membranes for the hydrogen separation and purification, since they combine the benefits of the two aforementioned inorganic membrane typologies. Indeed, the deposition of dense Pd-layer onto a porous support presents advantages like high H_2 perm-selectivity, high permeability, and lower presence of palladium. The supports useful for the Pd-based composite membranes preparation can be based on porous stainless steel (PSS), which are mechanically stable and easy of allocating in a module, or on porous ceramic and Vycor glass supports, which suffer the thermal cycles more than the metallic supports owing to the larger difference of the dilation coefficient between palladium and the ceramic material.

As a consequence, a PSS-supported Pd-based membrane can offer an optimal compromise among moderate cost, high H_2 permeability and perm-selectivity, and good mechanical resistance. Nevertheless, they are characterized by a large pore distribution, which can cause the intermetallic diffusion of palladium, decreasing the permeation characteristic of the membrane. Therefore, the PSS supports are previously conditioned with an intermediate layer useful to decrease their pores size.

Usually, different techniques can be used for depositing a thin selective Pd-layer and, depending on the technique, there is a threshold on the minimum thickness obtainable by the deposition method. This limit increases with the pore size of the support.

However, composite Pd-based membranes can be characterized by a series of issues related to inherent instability and caused by the different nature of the materials. One of them is represented by the membrane instability to the interface Pd-layer/support, i.e., the adherence loss, flaking-off, and cracking, potentially caused by the different thermal expansion coefficients and by the selective layer volume variation, due to the hydrogen diffusion in the palladium lattice. Furthermore, other forms of instability are connected to Pd-based membrane micro-structure, generated by the interaction of the selective layer with the support and the gaseous environment. As a consequence, thin membranes are characterized by very small grains and high density of grain borders, which could lead to grain size growing, impurities dissolution, grain borders diffusion, and alloy segregation.

The main effects of these instabilities are represented by a reduction of the membrane performance and this effect is more evident at high temperature.

The composite Pd-based membranes are commonly not fully hydrogen perm-selective and the hydrogen permeating flux can be determined by using Eq. (7) as reported in the following:

$$J_{H_2} = P_{H_2}\left(p^n{}_{H_2,retentate} - p^n{}_{H_2,permeate}\right), \tag{7}$$

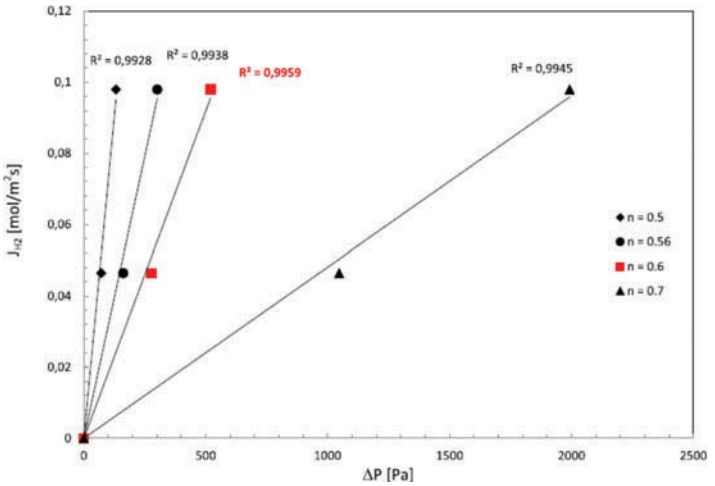


Figure 4. Evaluation method of "n" factor for a supported Pd-based membrane based on the experimental hydrogen permeating flux vs. transmembrane pressure. Reprinted from Iulianelli, et al. [67], with permission of reprint from Elsevier Ltd.

where P_{H2} is the hydrogen permeance and n the dependence factor of the hydrogen flux to the hydrogen partial pressure, which can vary from 0.5 (used for indicating Sieverts-Fick's law and applied for dense Pd-membranes) to 1 (used when, at high pressures, the hydrogen-hydrogen interactions in the palladium bulk are not negligible). In Fig. 4, it is illustrated schematically how "n" factor is calculated for a supported Pd-based membrane. In details, a plot of the experimental hydrogen permeating flux versus the transmembrane pressure at different "n" is drawn and a linear regression equation with associated R^2 is used for each different "n" The "n" factor is, then, chosen when linear regression shows the maximum R^2 factor.

However, both dense self-supported and composite Pd-based membranes have the peculiarity to be hydrogen perm-selective with respect to all the other gases. Therefore, both kinds of membranes, when allocated in MRs, allow for overcoming the thermodynamic restrictions due to the removal of hydrogen from the reaction side through the membrane ("shift effect"). For the Le Chatelier principle, the reaction shifts towards the product, with a consequent enhancement of the conversion and with the further advantage of collecting high-grade hydrogen in the permeate side of the MR.

4. Steam reforming of methane for hydrogen generation and membrane reactors

The most severe limitation of MSR reaction performed in CRs is represented by the thermodynamic constraints, which impose harsh operating temperature and pressure to reach high methane conversion. As a consequence,

under these severe conditions the catalysts could be deactivated due to carbon formation, which could cause some blocks, responsible for pressure drops along the reactor. Therefore, with the aim of depleting the high energy requirements and avoiding the catalyst fouling, the MR technology could be adopted, meanwhile producing different benefits such as the generation of highly concentrated hydrogen. Furthermore, many scientists demonstrated that the MRs can promote realistic enhancements of MSR reforming process in terms of costs reduction and milder operating conditions.

Then, in the contest of MR technology, two different kinds of reforming reactors are allowed: packed-bed membrane reactors (PBMRs) and fluidized-bed membrane reactors (FBMRs).

PBMRs are the most studied and they can be distinguished in two distinct configurations such as tubular or plane, with the catalyst loaded inside to constitute a packed bed in direct contact with the membrane (108). In tubular configuration, Fig. 5, the catalyst can be loaded either in the membrane core or in the annulus.

The PBMRs show a simple configuration, in which catalyst particles can be loaded and their dimension plays an important role, being responsible for the performance of this reactor. Small particles can increase the pressure drops, whereas big particles can affect negatively the system in terms of internal mass transfer. Particularly relevant for the PBMRs is the mass transfer limitation from the bed to the wall, causing reduced hydrogen permeation and remarkable temperature profile along the reactor, with consequent negative effects on catalyst and membrane performance (109).

Concerning the FBMR technology, in a pioneeristic study of Adris et al. (110), a common FBMR configuration is represented by a bundle of membranes immersed in a catalytic bed, exercised at bubbling or turbulent regime. Compared to the PBMRs, the FBMRs show higher catalyst effectiveness and improved mass and heat transfer rates, with consequent more uniform temperature profile along the reactor. Otherwise, the minimum

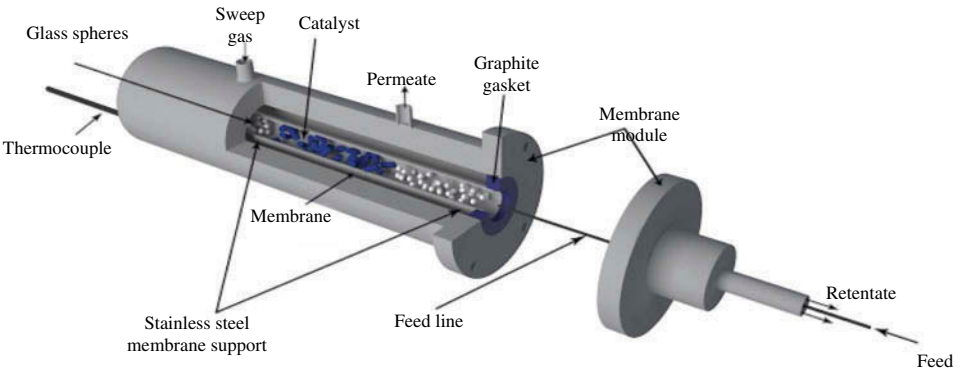


Figure 5. Scheme of a packed-bed membrane reactor in tubular configuration.

fluidization velocity represents the limit of this kind of reactors. Indeed, the catalytic bed is in suspension only in case the minimum fluidization velocity is overcome. Nevertheless, the presence of bubbles makes possible some mass transfer limitations, negatively affecting the methane conversion during MSR reaction.

4.1. *Hydrogen generation during MSR reaction in PBMRs and FBMRs*

In the last decades, several scientific papers dealt with the MSR reaction to generate hydrogen in PBMRs, equipped with palladium or palladium-alloyed membranes, although in recent years most attention has been dedicated to the application of the FBMRs for producing hydrogen. As a small overview, Table 4 resumes some representative scientific results regarding MSR reaction in both PBMRs and FBMRs. In detail, in the contest of MSR reaction to generate highly concentrated hydrogen, the first application of a PBMR in flat configuration (allocating a Pd disk of 100 μm of thickness) was realized in 1987 by Oertel et al. (111), demonstrating superior performance in terms of methane conversion with respect to a CR.

Successively, Uemiya et al. (112) performed MSR reaction at medium temperature (500°C) and 9.1 bar in a composite Pd-based PBMR (Pd deposited onto a porous glass support) reaching 90% methane conversion. Other authors performed MSR reaction at medium temperature even though there is a discrepancy among experimental results in literature, probably due to the utilization of different catalysts and/or operating conditions such as pressure, steam to carbon feed ratio etc.

For example, Shu et al. (113) achieved 80% methane conversion at 500°C, by using a lower reaction pressure, 2.0 bar, with respect to the work of Uemiya et al. (112), by using a PBMR housing a composite membrane constituted of a Pd-Ag layer deposited onto a super alloy support based on Ni-Cr. Otherwise, at the same temperature conditions and 6.0 bar, Jorgensen et al. (114) reached 51% of methane conversion by using a PBMR allocating a thick Pd-Ag membrane (100 μm). At 500°C, Kikuchi et al. (115) obtained 100% methane conversion by operating at ambient pressure in a PBMR housing a composite Pd-based membrane. At the same temperature and by using high pressure (20.0 bar), Lin et al. (28) reached almost 90% methane conversion, but recovering *ca* 90% of the produced hydrogen in the permeate side. The authors also pointed out the benefit of the PBMR in terms of energy saving comparing the results with an equivalent CR, which can achieve the same conversion working at 850°C.

At higher temperature, ~550°C and 3.0 bar, Tong et al. (30) reached 97% methane conversion and more than 90% hydrogen recovery by means of a PBMR allocating a supported PSS Pd-based membrane. At the same temperature and a higher pressure (9.0 bar), a composite membrane constituted

Table 4. Experimental data from the open literature about MSR reaction in PBMRs and FBMRs.

Housed Membrane	Reactor configuration	Catalyst	T [°C]	Pressure [bar]	Hydrogen recovery [%]	CH_4 conversion [%]	Reference
Flat Pd-dense	PBMR	Ni-based	550	–	–	> 80	Oertel et al. (111)
Pd-supported onto glass	PBMR	–	500	9.1	–	90	Uemiya et al. (112)
Pd-Ag supported onto a super alloy	PBMR	Ni/Al_2O_3	500	2.0	–	80	Shu et al. (113)
Pd-Ag supported onto PSS	PBMR	Cu/ZnO	500	6.0	–	50	Jorgensen et al. (114)
Pd-based	PBMR	Ni/Al_2O_3	500	1.0	–	100	Kikuchi et al. (115)
Pd-supported onto PSS	PBMR	Ni/Al_2O_3	500	20.0	90	86	Lin et al. (28)
Pd-supported onto PSS	PBMR	Ni/Al_2O_3	550	3.0	91	97	Tong et al. (30)
Pd-supported onto Al_2O_3	PBMR	Ni-La/Mg-Al	550	9.0	95	˜ 100	Chen et al. (22)
Dense self-supported Pd-Ag	PBMR	Ni/Al_2O_3	450	3.0	70	50	Iulianelli et al. (24)
Dense self-supported Pd-Ag	PBMR	Ni/ZrO	450	5.0	80	65	Basile et al. (21)
Pd-supported onto Al_2O_3	PBMR	Ni-based	580	28	66	98	Sarıc et al. (27)
Pd-supported onto ceramic	PBMR	Ni-based	650	16	65	70	Dittmar et al. (20)
Pd-supported onto alumina	PBMR	$Pt\text{-}Ni/CeO_2$	525	10	> 80	90	Patrascu & Sheintuch (116)
Pd-supported onto dense ceramic	PBMR	Ni-based	550	1	–	˜ 70	Kyriakides et al. (117)
Dense Pd-tube	FBMR	–	640	10	> 40	˜ 100	Adris et al. (120)
Pd supported onto a super alloy	FBMR	–	650	4.0	–	97	Patil et al. (129)

by a thin Pd-layer onto alumina and allocated in a PBMR module was used by Chen et al. (22), reaching almost complete methane conversion over around 30% of an equivalent CR, with the further advantage of collecting 95% hydrogen in the permeate side.

In the study of Iulianelli et al. (24), a dense self-supported Pd-Ag membrane (50 μm thick) allocated in a PBMR allowed for 50% methane conversion and 70% hydrogen recovery at 450°C and 3.0 bar. In this case, low Ni concentration in the used catalyst (wt 0.5%) was probably the main reason of the low performance in terms of methane conversion, which was improved in a following study reaching around 65% by using Ni/ZrO_2 catalyst and 5.0 bar of reaction pressure (21).

In most recent years, an interesting literature on the recent advances about MSR reaction to produce hydrogen in PBMRs can be found. For example, Saric et al. (27) paid special attention not only to achieve the best experimental results in terms of conversion and hydrogen recovery (98% methane conversion and around 70% hydrogen recovery at 580°C, 28 bar, and 3/1 as steam to carbon feed ratio), but also to investigate the resistance of the composite Pd-based membrane under operation in order to give more details to transfer this technology from bench scale to pilot or industrial scale. In particular, they observed more than 1100 h as life time of the membrane at stable permeation characteristics.

In the same contest, Dittmar et al. (20) studied the long-life time of a Pd-based composite membrane under operation at different temperatures, demonstrating that the membrane performance are stable at 650°C, meanwhile reaching 70% methane conversion and 65% hydrogen recovery at 16 bar of reaction pressure.

Iulianelli et al. (67) used a model biogas mixture for MSR reaction performed in a composite Pd/Al_2O_3 MR to generate renewably and clean hydrogen for fuel cells applications. The composite membrane showed H_2/N_2 ideal selectivity equal to 4300 at 0.5 bar of transmembrane pressure and T = 400°C. When allocated in the MR module, this solution made possible a methane conversion higher than 30% at 450°C, 3.5 bar $H_2O:CH_4$ = 4:1, GHSV = 11000 h^{-1} with a quite high hydrogen recovery around 70%. Nevertheless, at the aforementioned operating conditions, the characteristics of H_2 perm-selectivity were negatively affected causing an increased presence of pin-holes formation on the Pd-layer of the composite membrane and inducing a H_2 permeate purity lower than 70%. Otherwise, at 380°C, 2.0 bar, $H_2O:CH_4$ = 3:1, GHSV = 9000 h^{-1} although the MR reached a modest conversion (27%), the hydrogen recovered in the permeate side (> 20%) showed a purity around 96%.

Regarding the FBMRs, Adris et al. (118,119) were the first scientists involved in the development of this kind of MR technology. In particular, the authors studied MSR reaction in a FBMR from a modeling point of view,

demonstrating that complete methane conversion could be reached by inserting in the model the typical industrial operating conditions. Successively, the same authors validated the model prediction, by performing experimental tests (120). At 640°C and 10 bar, they achieved 100% methane conversion with a hydrogen recovery higher than 40%; see Table 4. Over the years now, a number of other researchers investigated MSR reaction in FBMR from both modeling (37,122–128) and experimental point of view, even though only few researchers spent their efforts on the experimental aspect probably because of the hurdles of the scaling up of this kind of reactor at larger scale, the difficulties in the reactor building up, membrane sealing, and erosion issues. (38,40,41,121,129). Among them, worth noting is the study of Patil et al. (129), who reached around 97% methane conversion, carrying out MSR in a FBMR at 650°C and 4.0 bar; see Table 4.

Therefore, most of the scientific articles on this kind of MR typology has been dedicated to really detailed modeling frameworks. Thus, a panoramic view about MSR reaction in both FBMR and PBMR from a theoretical perspective is analysed and reported in the next section.

5. Methane steam reforming kinetic

In the past, the investigations on MSR reaction have been mainly addressed towards the preparation of catalyst and the process analysis, neglecting both the kinetics and the reaction mechanisms with a consequent lack of information. Therefore, in recent decades, several studies have been realized with the intent to propose coherent reaction mechanism and kinetic data (77,83,130–135). One of the first accurate studies on MSR kinetics was realized by Temkin et al. (130), who conducted the experimental tests on nickel foil at atmospheric pressure and by varying temperature from 470–900°C. In particular, they discovered that, at the highest temperature, the reforming reaction followed a first-order equation and, at lower temperature, a delay in the rate of hydrogen formation was detected. Ten years later, Xu and Froment (131,132) improved the Temkin's model developing the most well-known kinetic model for MSR reaction. In particular, Xu and Froment initially provided a path of reactions consisting of 21 rate equations, which were reduced to three by considering the Langmuir-Hinshelwood reaction mechanism. So, they identified the three main well-known reactions (Eqs. (1)–(3)) and formulated the corresponding rate reaction expressions (Eqs. (8)–(10)):

$$r_1 = \frac{\frac{k_1}{p_{H_2}^{2.5}} \cdot \left[p_{CH_4} \cdot p_{H_2O} - \frac{p_{H_2}^3 \cdot p_{CO}}{K_{eq1}} \right]}{\left(1 + K_{CO} \cdot p_{CO} + K_{H_2} \cdot p_{H_2} + K_{CH_4} \cdot p_{CH_4} + K_{H_2O} \cdot \frac{p_{H_2O}}{p_{H_2}} \right)^2} \quad (8)$$

$$r_2 = \frac{\frac{k_2}{p_{H_2}} \cdot \left[p_{CO} \cdot p_{H_2O} - \frac{p_{H_2} \cdot p_{CO_2}}{K_{eq2}} \right]}{\left(1 + K_{CO} \cdot p_{CO} + K_{H_2} \cdot p_{H_2} + K_{CH_4} \cdot p_{CH_4} + K_{H_2O} \cdot \frac{p_{H_2O}}{p_{H_2}} \right)^2} \quad (9)$$

$$r_3 = \frac{\frac{k_3}{p_{H_2}^{3.5}} \cdot \left[p_{CH_4} \cdot p_{H_2O}^2 - \frac{p_{H_2}^4 \cdot p_{CO_2}}{K_{eq3}} \right]}{\left(1 + K_{CO} \cdot p_{CO} + K_{H_2} \cdot p_{H_2} + K_{CH_4} \cdot p_{CH_4} + K_{H_2O} \cdot \frac{p_{H_2O}}{p_{H_2}} \right)^2}, \quad (10)$$

where k_i are the rate coefficient of reactions, K_i adsorption constants, and $K_{eq,i}$ equilibrium constant of reactions.

Over the years now, other researchers, such as Oliveira et al. (133) and Wei et al. (83), followed the Xu and Froment's work, proposing their own MSR mechanism and investigating the reaction on a Ni-based catalyst. In particular, they focused their attention on the rate expression for CH_4 reforming reaction owing to the controversies and often contradictory kinetic developed. For instance, Wei et al. (83) performed MSR reaction on Ni/MgO catalyst at 550–750°C, revealing that the reaction rates were dependent on CH_4 decomposition and indicating that the C-H bond activation was the only determining step in all of the three reactions.

Tsipouriari et al. (136) also proposed a scheme in which the adsorption of CH_4 on Ni, followed by cracking and carbon deposition, was assumed as the slow step.

However, several studies on MSR kinetic were carried out by also using different catalysts, such as Rh and Pt based. As a result, different kinetic expressions were found. For instance, Maestri et al. (77) investigated the MSR catalytic mechanism on Rh/Al_2O_3 at 400, 500, and 600°C, disclosing that the methane activation was the rate limiting step and the steam, dissociating into H and OH, oxidized the carbon coming from the methane pyrolysis.

Schadel et al. (135) developed a kinetic model by considering the CH_4 dissociative adsorption as the limiting step. Moreover, they performed MSR reaction over Rh/ZrO_2 catalyst at 425–575°C, finding a good match between experimental results and kinetic model.

In summary, so far many studies have been devoted to the investigation of MSR kinetic, by analyzing different catalysts and supports. In most of them, different rate limiting steps and, consequently, different kinetic expressions have been found. According to Jones et al. (82) and Ligthart et al. (26,73), the

differences in kinetics mechanisms are caused by the diverse particle sizes, metal dispersion, and support used in each study.

5.1. *PBMR and FBMR: Modeling results*

As is well known, the modeling is an important tool for describing, optimizing and controlling complicated system. Currently, several mathematical models for MSR reaction performed in PBMRs and FBMRs are present in the specialized literature (137–159). They are based on different assumptions and reactor configurations and their complexity is a function of the number of conservation equations considered. Generally, the mass and energy balance as well as the continuity equation and momentum balance are required. The first two equations are used for predicting the variation of concentration and temperature; the last two ones for assessing the mass changes due to permeation through the membrane and the pressure variation.

Each reactor model can be classified according to its complexity. In particular, regarding the PBMRs the model can be categorized according to the number of dimensions, such as one-dimensional (1D), bi-dimensional (2D), or tri-dimensional models (3D). Meanwhile, for the FBMRs, two model classes have been proposed based on pseudo-homogeneous or two-phase approach, respectively (137).

Generally, 1D model takes into account only the gradient in the axial direction. This model is very simple to solve and it is frequently used (31,138–142). For instance, Simakov and Sheintuch (31) used a pseudo-homogeneous 1D model for optimizing, in terms of membrane area and heat losses, an autothermal MR to produce hydrogen via MSR reaction.

The 2D model is used when it is not possible neglecting the radial gradient, for instance, in the case of mass and heat transfer through the membrane (143–147). Although this model is more accurate, it involves more calculations and, consequently, it is high computing time-consuming. For example, Tiemersma et al. (146) used a CFD 2D model by highlighting the negative influence of the polarization effect on the membrane performance.

Markatos et al. (148) and Vogiatzis et al. (149) simulated MSR reaction at industrial level for investigating the mass effect and heat dispersion.

Oyama and Hacarlioglu (144), compared the PBMR performance, simulating with 1D and 2D models, and suggested a criterion for choosing which model should be used at the given operating conditions. An improvement of this work was realized by Marìn et al. (143), who also studied the effect of the operating conditions for optimizing the PBMR performance, highlighting that the space velocity and the reaction temperature can be considered as the most important parameters.

Roses et al. (159) developed a bi-dimensional CFD model to simulate a fuel processor for hydrogen production from natural gas or biogas based on a realistic architecture, by analyzing the feasibility of thermally integrating an endothermic PBMR with a stream of hot gas. The resulting fuel processor arrangement made possible to achieve a very high efficiency, particularly suited for integration with fuel cells.

Regarding the 3D models, they are complex and are implemented when no-symmetrical reactors need to be modeled (150,151).

However, over the years several researchers have investigated the modeling of PBMRs and FBMRs. Table 5 shows an overview of the mathematical models and the main results realized by simulating MSR reactions in both reactor typologies.

Regarding the FBMRs, the models can be classified according to the following approach (pseudo-homogeneous or two-phase one) (137).

The first approach proposes conventional multiphase flow models, such as ideal flow, dispersion, and residence time distribution models. The second approach considers two phases, such as bubble and emulsion, suggesting for each phase different governing equations and describing the mass interchange between them.

The "bubbling bed model" developed by Kunii and Levenspiel (156) and the "bubble assemblage model" proposed by Kato and Wen (157) are the most used two-phase models. In particular, in the first model the bubble phase flows in a plug flow in a well-mixed emulsion phase. In this case, the mass exchange

between the bubble and emulsion is taken into account by a constant average bubble diameter (39).

In the bubble assemblage mode, the fluidized bed is subdivided in "n sections," which are considered as ideally continuous stirred tank reactors.

However, both pseudo-homogeneous and the two-phase approaches have been used for modeling various FBMRs. For instance, Adris and Grace (120) proposed a combination between the bubbling bed model and the equation of the hydrogen permeation through the membrane for MSR reaction, showing a good match with their experimental data. Successively, Rakib and Alhumaizi (127) extended this model, highlighting the better performance in terms of hydrogen permeation reached in a FBMR by using coated membrane tubes.

Other researchers (158,41,40,153) considered the bubbling fluidized beds approach for modeling MSR reaction in FBMRs. Recently, Ye et al. (154) used Aspen Plus integrated with a Fortran sub-routine for simulating this reaction, showing a good agreement between the model predictions and the experimental data taken from literature. Moreover, Saebea et al. (155) proposed FBMR and CR models for simulating biogas steam reforming reaction by using different operating conditions. They showed that the FBMR allows

Table 5. Overview on mathematical models for simulating MSR reaction by using PBMRs and FBMRs.

Reactor configuration	Classification	Assumption	Results	References
PBMR	1D	Pseudo-homogeneous membrane full H_2 perm-selective isothermal conditions Xu and Froment (131)	Optimized MSR reaction in terms of membrane area and heat losses	Simakov and Sheintuch (31)
PBMR	1D	No-isothermal conditions plug flow membrane full H_2 perm-selective Ergun's equation Xu and Froment (131)	The distribution of reactive/permeating stage can be neglected	Caravella et al. (138)
PBMR	1D	No-isothermal conditions membrane full H_2 perm-selective Ergun's equation	Membrane performance affected by hydrogen diffusion in case of thickness falling in a particular region	Huppmeier et al. (139)
PBMR	1D	Pseudo-homogeneous plug flow membrane full H_2 perm-selective Ergun's equation	Complete CH_4 conversion and relatively high H_2 recovery for PBMRs smaller in dimensions than CRs	Patel and Sunol (140)
PBMR	2D	Pseudo-homogeneous plug flow membrane full H_2 perm-selective Xu and Froment (131)	Study about reaction temperature and space velocity as the most important parameter affecting the PBMR performance	Marìn et al. (143)
PBMR	2D	Pseudo-homogeneous membrane full H_2 perm-selective no-isothermal conditions Xu and Froment (131)	Study about the efforts addressed towards thermal stability of the membrane and its permeability in a PBMR	De Falco et al. (145)
PBMR	2D	Pseudo-homogeneous membrane full H_2 perm-selective	The effect of concentration polarization is emphasized by high H_2 permeability. The porosity profile needs to be considered in order to avoid the hydrogen permeating overestimation	Tiemersma et al. (146)
PBMR	1D	Pseudo-homogeneous plug flow membrane full H_2 perm-selective Xu and Froment (131)	Good prediction of PBMR performance in terms of composition and permeate flow	Patrascu and Sheintuch (152)
PBMR	2D	Bi-dimensional CFD simulation models a system based on a realistic architecture, properties for gas mixtures are calculated with ideal gas mixing law membrane full H_2 perm-selective	The CFD simulation confirmed the feasibility of thermally coupling a membrane reformer for hydrogen production with the hot exhaust gasesavailable in an architecture intended for use in a PEMFC-based micro-CHP system.	Roses et al. (159)

(*Continued*)

Table 5. (Continued).

Reactor configuration	Classification	Assumption	Results	References
FBMR	1D	Membrane full H_2 perm-selective dense catalytic bed composed by two phases, a bubble phase and a dense phase. reaction occurs in the dense phase	The membrane capacity and it location between catalyst and free board region have been considered as important design parameter in this configuration	Adris et al. (118)
FBMR	–	Three models based on different assumption: bubbling regime turbulent regime fast fluidization	The best FBMR performance have been reached by considering the turbulent regime.	Andrès et al. (153)
FBMR	1D	Membrane full H_2 perm-selective uniform temperature pressure gradients are ignored	The simulation has been carried out with Aspen Plus showing a good agreement with the results in the literature	Ye et al. (154)
FBMR	1D	Membrane full H_2 perm-selective uniform temperature pressure gradients are neglected steady state	FBMR reaches considerably high hydrogen production by carrying out biogas steam reforming reaction, compared with the CR.	Saebea et al. (155)

for better performance in terms of hydrogen production compared to the equivalent CR.

In summary, it is noteworthy that a single model valid for performing MSR process in PBMRs and FBMRs has been not yet developed and that further investigations are required for increasing the knowledge in the modeling of such reactors.

6. Conclusion

This article pointed out that MSR reaction represents a mature process not only because it was carried out in conventional reactors but also because it involved a strong interest in developing alternative tools and more active catalysts for energy and costs saving. Furthermore, this study has also reported a short overview on the most recent advances about catalysts utilization in MSR reaction to generate hydrogen. In addition, the role of the support has been also discussed in the field of coke formation reduction and catalytic activity enhancement.

Thus, this article also presented the recent advances about the MR technology, analyzing the different solutions for enhancing the MSR process. In particular, the typical benefits of the MRs have been discussed considering the synergistic combination of two process functionalities in a single unit (the MR), by subdividing the discussion between PBMRs and FBMRs applications. In particular, the functionalities of PBMRs, characterized by very simple configuration have been illustrated as well as the FBMRs, showing enhanced mass and heat transfer rates and favouring more uniform temperature profiles. Nevertheless, for the latter reactor typology, possible bubble-to-emulsion mass transfer limitations can negatively affect the FBMRs performance and the catalyst fluidization can damage the membrane.

Then, MSR reaction in MRs has been also reviewed to offer a panoramic view on the evolution of the theoretical approach to this process and the recent efforts.

In conclusion, it can be stated that the progress realized until now on MSR reaction in MRs suggests that other further challenges for MRs commercialization will be met in the future. Nevertheless, from an industrial point of view there are still many issues to be addressed, such as the costs of the membranes and their stability with time on stream.

Acronyms

1D	One-dimensional
2D	Bi-dimensional
3D	Tri-dimensional
CFD	Computational fluid dynamic

CR	Conventional reactor
FBMR	Fluidized bed membrane reactor
FBR	Fixed bed reactor
GHG	Green-house gases
HTS	High temperature shift (reactor for water gas shift carried out at high temperature)
LTS	Low temperature shift (reactor for water gas shift carried out at low temperature)
MR	Micro membrane reactor
MSR	Methane steam reforming
PBMR	Packed bed membrane reactor
PEMFC	Proton exchange membrane fuel cell
PrOX	Preferential oxidation
PSA	Pressure swing adsorption
PSS	Porous stainless steel
WGS	Water gas shift

Symbols

r	reaction rate
p	partial pressure
k	rate coefficient of reactions
K	adsorption constant
K_{eq}	equilibrium constant of reactions
E_a	apparent activation energy
H	permeation rate
J_{H2}	hydrogen flux permeating through the membrane
n	dependence factor of the hydrogen flux to the hydrogen partial pressure
Pe^0	pre-exponential factor
P_{H2}	hydrogen permeance
$p_{H2\text{-}permeate}$	hydrogen partial pressures in the permeate side
$p_{H2\text{-}retentate}$	hydrogen partial pressures in the retentate side
R	universal gas constant
T	absolute temperature
δ	membrane thickness

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